

Since GPUs continuously generate large amounts of waste heat during AI training, could electric/GPU stoves utilize this unused thermal energy resource through on-demand tickets as distributed networks instead of citizens using a wasteful private electric stove? What are the scientific challenges?

Posted by u/Few-Elephant-2600 • 0 points • 17 comments

What if you could cook dinner using the excess heat from your GPU stove that you rent out to others for AI training? Could distributed GPU stove systems help optimise energy use in the globe with cooking your dinner while powering another's tasks?

Comments

u/Eggman8728 • 1 points • 2024-03-12 13:10 UTC

First of all, you'd need to concentrate all the heat from those GPUs into the inside of the oven while keeping them cool. That would be very energy inefficient. Then you've got the problem of cost. A 4090 costs about 1500\$, while drawing 450 watts of heat. Assuming you somehow get all that heat into the inside of your oven, you'd want about four of them to match the output of a normal oven heating element. Congrats, your oven now costs more than 6000\$.

u/bree_dev • 1 points • 2024-03-12 06:43 UTC

You can't mix three things at 70C to get 210C.

u/ChipotleMayoFusion • 1 points • 2024-03-12 06:14 UTC

The main problem is the heat tolerance of computer parts, most of which are intended to operate in an ambient temperature up to 75F or 25C.

An oven is an insulated box that you heat to cook food. If you want to power an oven with a computer, the computer needs to be inside the oven insulation, otherwise the heat will just go into the room instead of building up and cooking the food.

A stove top has a similar problem, the idea is you release heat below a pot or pan to cook the food inside. If you want to boil water you need at least 212F or more to handle losses.

It is possible to build some computer parts in a way that can handle 350 F, but some of them would be beyond the current level of technology.

u/eliminate1337 • 1 points • 2024-03-12 05:11 UTC

There's no scientific challenge. It's a matter of if the energy savings are worth the cost.

The silicon die itself should stay below 100C. The exhaust air is cooler, maybe 60C. Not hot enough for cooking but possibly useful for water heating. It's just not worth the cost to put thousands of dollars of computers in a stove that's on for a fraction of the day.

u/Few-Elephant-2600 • 1 points • 2024-03-12 05:28 UTC

Can't the cost be brought down to the level of Nvidia's next GPU, which will hopefully have extra features to make it useable as a frying pan and have extra heat resistance?

u/AlphaState • 2 points • 2024-03-12 04:54 UTC

Processors can't run at the temperature of a stove, I don't think any silicon could. The problem with waste heat is that you can't just ramp the temperature (60-70C maybe) up to another temperature (180C) so it doesn't help much.

My PC does help keep my home office warm in winter though.

u/Jofarin • 1 points • 2024-03-12 07:34 UTC

You can't? Isn't that how compressor based fridges work? Cool down a coolant to the inside temperature, ramp it up a lot by compressing the air so it's above outside temperature, cool down the coolant by exposing it to outside temperature, cool down more by decompressing, rinse repeat.

Is there a fundamental misunderstanding on my side?

u/ccdy • 1 points • 2024-03-12 13:41 UTC

Yes, there is. Heat pumps are just heat engines run in reverse. Instead of extracting work from a temperature difference, they do work to create a temperature difference. A GPU does no work (apart from the cooling fans) and only produces heat.

u/Jofarin • 1 points • 2024-03-12 14:17 UTC

But you could use a heat pump to use the heat produced by a GPU to heat up a stove is what I'm saying.

u/AlphaState • 1 points • 2024-03-12 07:39 UTC

Of course you can, but you have to put more energy in. The temperature difference of a processor from ambient is small so it doesn't contribute much.

u/Jofarin • 1 points • 2024-03-12 07:46 UTC

It's a couple hundred watts, which isn't too much given stoves have like 2 kilowatts I think.

But good, ai was just confused by the statement it wouldn't be possible.

u/arsenic_kitchen • 6 points • 2024-03-12 04:24 UTC

Electric stoves aren't inefficient. They convert electricity to heat very efficiently. That's what they're built to do.

GPUs are inefficient.

u/Few-Elephant-2600 • 1 points • 2024-03-12 04:57 UTC

Yes! That's why I choose it actually. I guess GPUs essentially resistors that emit heat and change bits in the process. Wouldn't it be clean job to provide all the processing power the world needs, not just for AI, but for all CPU and cloud processes, while cooking?

u/mfb- • 1 points • 2024-03-12 05:53 UTC

- * A GPU that's only getting used for a few minutes per day is a waste of money.
- * GPUs don't reach the typical temperatures you want for cooking, so you still need some additional heat source.

Heating a home is a better application, but then you still have to work out how to make that work in practice. Who owns the GPU? The home owner or some external company? Who pays for the electricity, and if it's not the home owner who measures how much of the house electricity is used for the GPU?

u/arsenic_kitchen • 1 points • 2024-03-12 05:05 UTC

I remember reading about a scheme intended to capture waste heat from server farms *years* ago... as with so many things, it would benefit society, but cost money to build the infrastructure. A personal home GPU isn't wasting nearly so much heat as farms. Although, I don't mind abusing my GPU in the winter for a little bonus space heater.

u/rddman • 1 points • 2024-03-12 04:18 UTC

I think the challenge would be to remove heat from the GPU and then instead of dissipating it as usual, somehow accumulate it to heat a cooking plate to sufficient high temperature. For cooking you need a temperature of at least 100c, but a GPU cooler at that temp would normally mean the GPU itself stays way to hot.

u/billndotnet • 14 points • 2024-03-12 04:12 UTC

Yes but no but yes.

Cooking is probably the wrong application, because it's high output and relatively short term.

What you're probably looking for is room, whole home, or hot water heating, instead.

<https://www.bbc.com/future/article/20171013-where-data-centres-store-info---and-heat-homes>